

Mihir Ravindra Adelkar

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Summary

Software Engineer with 3+ years building scalable microservices and distributed systems. Expert in Node.js, Python and Java with deep experience in message queues (Kafka), API design (REST/GraphQL/gRPC), and cloud-native deployments. Proven track record processing 1B+ daily messages, reducing latency by 70%, and achieving 99.9% uptime through robust architecture and observability.

Skills

Languages	Python, Java, Go, TypeScript, JavaScript, SQL, Bash/Shell, C++
Backend	Node.js (Express/NestJS), Python (Django/Flask/FastAPI), Java (Spring Boot), Go, RESTful APIs, GraphQL, gRPC
Messaging	Kafka (producers/consumers, exactly-once, DLQ), RabbitMQ, Redis Pub/Sub, Event-Driven Architecture
Databases	PostgreSQL, MySQL, MongoDB, Redis, DynamoDB; Schema Design, Indexing, Query Optimization, Migrations
Cloud & DevOps	AWS (EKS, Lambda, RDS, S3, SQS), GCP, Docker, Kubernetes, Terraform, Istio, Jenkins, GitHub Actions
Observability	Prometheus, Grafana, New Relic, OpenTelemetry, K6 Load Testing, SonarCloud, Distributed Tracing

Professional Experience

Backend Developer, KeelWorks — Boston, MA

Aug 2025 – Present

- Building backend services in **Node.js** with **PostgreSQL**; implementing **Kafka-driven** async workflows and **Redis caching**.
- Designing data models and optimizing queries for workforce engagement platform serving distributed teams.
- Deploying containerized microservices on **AWS EKS** with Docker/Terraform; implementing **CI/CD** pipelines.

Software Engineer, Freelancing (Dabadu.ai) — Remote

May 2023 – Sept 2023

- Stabilized backend services for automotive dealership CRM; implemented structured logging and error handling.
- Integrated lender APIs into backend workflows with **Node.js + PostgreSQL**, adding retry logic and circuit breakers.
- Established **SonarCloud** quality gates and automated testing, reducing production bugs by **20%**.

Software Engineer, NeoSOFT Technologies — Mumbai, India

Jun 2021 – May 2023

- Architected scalable backend services using **Node.js**, **Django**, and **Spring Boot** deployed on AWS/GCP, serving **500K+ users**.
- Led migration of monolithic application into **Docker containerized microservices**, reducing deployment time by **80%**.
- Designed **API gateway** patterns with rate limiting, request validation, and circuit breakers for fault tolerance.
- Implemented **Kafka** for async processing handling **1B+ messages daily** with exactly-once semantics and DLQ patterns.
- Designed complex **PostgreSQL** schemas with normalization, indexing strategies, and **Liquibase/Flyway** migrations.
- Designed **RESTful and GraphQL APIs** with auth, versioning, pagination; added **gRPC** reducing cross-service latency by **30%**.
- Optimized **PostgreSQL/MongoDB** queries and added **Redis caching**, reducing API response times by **70%**.
- Created reusable **Terraform modules** for RDS, ElastiCache, and EKS clusters across multiple environments.
- Added observability with **Prometheus/Grafana** and **New Relic**, reducing MTTR from 25 to 15 minutes.
- Implemented **OAuth2/JWT** authentication with refresh token rotation and token revocation.
- Mentored 8 junior developers** through code reviews, system design discussions, and Agile sprints.

Projects

CVE Intelligence System — Distributed Data Pipeline

Go, Node.js, PostgreSQL, Kafka, AWS EKS, Terraform, Istio

- Built distributed **microservices** in **Go** for real-time CVE data processing using **Kafka** message streaming.
- Designed **PostgreSQL** schemas with indexing; implemented **Istio service mesh** for secure inter-service communication.
- Automated infrastructure with **Terraform**; established monitoring with **Prometheus/Grafana** and distributed tracing.

Multi-Tenant CRM Backend (Wasalt/Nesto)

Spring Boot, Node.js, PostgreSQL, Kafka, Redis, gRPC, Kubernetes

- Architected multi-tenant backend with **Spring Boot/Node** microservices supporting **50+ roles** with tenant isolation.
- Implemented **Kafka event streaming** for notifications and **gRPC** for low-latency inter-service calls.
- Designed **PostgreSQL** schemas with row-level security; added **Redis caching** achieving **sub-100ms** responses.

Healthcare Prior Authorization System

Python, FastAPI, PostgreSQL, Redis, Claude API, FHIR

- Built **FastAPI** backend transforming 3-5 day manual reviews into automated **5-second decisions**.
- Designed clinical data pipeline with **FHIR** resource mapping, extracting structured medical data with **95%+ accuracy**.
- Implemented **Redis** caching for LLM responses and async task queues for document processing.

Pedgog — Analytics Backend

Django, PostgreSQL, Redis, Kubernetes, New Relic, K6

- Built **Django** REST APIs for real-time classroom analytics serving **1,500+ concurrent users** with **Redis caching**.
- Conducted **K6 load testing** and implemented auto-scaling policies; instrumented with **New Relic APM**.

Open Source Contributions

GitHub: NeoSOFTTechnologies

- Multi-Tenancy Admin Console**: Node.js/NestJS backend with RBAC, Keycloak SSO; Docker containerized with CI/CD.
- Microservices Boilerplate**: Express/NestJS templates with auth, logging, health checks; adopted by 15+ teams.

Education & Certifications

Master of Science in Software Engineering Systems — Northeastern University, Boston, MA

Sept 2023 - May 2025

Bachelor of Engineering in Computer Engineering — University of Mumbai, Mumbai, India

Aug 2018 - Jun 2021

Certifications: Oracle Certified Java Enterprise Applications Developer | Scrum Fundamentals Certified (SCRUMStudy)